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A Report on

Smashifix : An Application for Badminton Posture Correction Using Deep Learning

Submitted in partial fulfillment of the requirement of the degree of

Bachelor of Technology in
Department of Computer Science and Engineering(Data Science)

By

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Under the guidance of

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CERTIFICATE

This is to certify that the project entitled, **Smashifix : An Application for Badminton Posture Correction Using Deep Learning** is a bonafide work of **Sian Rodrigues (60009230197), Vedant Gadge (60009230120), Yati Rathod (60009230026) and Smayan Kulkarni (60009230142)** submitted in the partial fulfillment of the requirement for the award of the Bachelor of Technology in Computer Science and Engineering (Data Science)

Prof. Pradnya Saval / Dr. Namita Pulgam

Head of the Department **Principal**

Place: Mumbai

Date: 18th October 2025

DECLARATION

We declare that this written submission represents our ideas in our own words and where others ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that, we have adhered to all the principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the Institute and can also evoke penal action from the sources, which have thus not been properly cited or from whom proper permission has not been taken, when needed.

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Place: Mumbai

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APPROVAL SHEET

Project entitled, **Smashifix : An Application for Badminton Posture Correction Using Deep Learning** submitted by **Sian Rodrigues (60009230197), Vedant Gadge (60009230120), Yati Rathod (60009230026) and Smayan Kulkarni (60009230142)** is approved for the award of the Bachelor of Technology in Computer Science and Engineering (Data Science).

Signature of Internal Examiner **Signature of External Examiner**

Place: Mumbai

Date: 18th October 2025

Content

1 Introduction

1.1 Overview

Computer vision and artificial intelligence have fundamentally transformed the landscape of sports analysis. Insights into player performance and biomechanics have reached unprecedented precision, enabling objective, data-driven evaluations. As discussed in **[14]**, modern computer vision systems possess the capability to track intricate patterns of movement across multiple sports, empowering coaches and athletes to identify areas for improvement through quantitative feedback. The limitations of human observation are now effectively bridged by automated systems, which provide real-time analysis and feedback with remarkable accuracy, marking a significant technological leap in athletic performance assessment.

1.2 Motivation

In badminton, a sport characterized by rapid gameplay and complex biomechanics, the application of computer vision is particularly impactful. The game demands exceptional precision in body mechanics, including footwork, stroke execution, and tactical positioning, making accurate movement analysis essential for player development. As highlighted in **[11]** and **[12]**, pose estimation algorithms and real-time perception pipelines have demonstrated strong potential for biomechanical tracking and feedback generation.

Computer-vision-based automated systems can analyze these movements in granular detail, often surpassing human capability by processing frame-by-frame motion data. Techniques such as part affinity field–based pose estimation **[11]** and trajectory-based player movement analysis have emerged as powerful methods for tactical evaluation and stroke categorization. These developments collectively underline the need for continued research and refinement of automated analysis systems, paving the way toward a future where sports training and competition can be further enhanced by computer vision innovations.

2 Literature Survey

2.1 Related work

Chun-Yi Wang et al explained a structured approach in **[1]** to data processing and annotation to create a high-quality dataset tailored for human pose estimation in sports. The dataset was built from high-definition videos recorded at 1920×1080 resolution, capturing a tennis player performing four key actions: forehand shot, backhand shot, serve, and ready position. From these recordings, frames were carefully extracted to ensure diverse pose representation, with each action category contributing 500 images, resulting in a total of 2,000 annotated frames.For the annotation process, the researchers used COCO-Annotator, an open-source tool that provides an interface for manually labeling human skeletal keypoints. Each image was annotated with 17 body keypoints in the COCO format, enabling compatibility with popular pose estimation models such as OpenPose, HRNet, and PoseNet. The annotated data was exported as JSON files adhering to the COCO standard, making the dataset directly usable in training or evaluating deep learning models. Technologies like Python, OpenCV, and COCO-Annotator played a significant role in preprocessing, frame extraction, and annotation. This methodical processing pipeline ensures that the dataset serves as a reliable benchmark for pose estimation tasks in dynamic, real-world sports environments like tennis.

Zhang Xipeng et al explores the integration of human pose estimation algorithms into badminton instruction to enhance teaching effectiveness. The authors developed a human-computer interactive teaching system in **[2]** that utilizes computer vision technology and neural network algorithms to analyse and compare students' movements against standard badminton techniques. This system aims to provide students with a more intuitive understanding of proper movements and hitting points, thereby improving their skills. The core of their approach is a human pose estimation method based on part affinity fields (PAFs). This method maintains the correlation of position and orientation information between limb regions, localizes key points of limb poses using part confidence maps, and integrates the PAFs to correlate all acquired feature key points, resulting in accurate human pose estimation. To train and evaluate this system, the researchers compiled a dataset comprising 30,000 training movements and 5,000 test movements, obtained from badminton match footage provided by relevant authorities. They manually classified and annotated these movements, focusing on key badminton skills such as forehand and backhand strokes

Gaetano Dibenedetto et al provides a comprehensive overview of deep learning-based Human Pose Estimation (HPE) in **[9]**, beginning with an introduction to its applications and the fundamental skeleton-based, surface-based, and volume-based human body models[1111]. It systematically categorizes and details various 2D and 3D HPE techniques, explaining the distinctions between regression-based and detection-based methods for single-person 2D HPE, as well as top-down and bottom-up approaches for multi-person scenarios. For 3D HPE, the paper organizes methods based on the input data, covering single monocular images, multi-view images, and video sequences[3]. Furthermore, the review explores non-camera-based HPE using technologies like radar and WiFi, lists essential frameworks and tools such as MediaPipe and OpenPose, describes key datasets like COCO and Human3.6M, and explains standard evaluation metrics like Average Precision (AP) and Mean Per Joint Position Error (MPJPE). The paper concludes by summarizing the performance of state-of-the-art models and highlighting ongoing research challenges, including occlusion, crowd detection, and real-time processing.

Qingyu Wang et al presents a deep learning-based framework for recognizing and correcting human posture in physical training using an improved convolutional neural network (CNN) architecture. At its core is a stacked hourglass network, which is particularly effective for human pose estimation due to its symmetrical structure that repeatedly compresses and expands feature maps, capturing both global and local contextual information. To enhance this architecture, the authors in **[3]** incorporate three key modules: the Improved Module, which addresses gradient vanishing issues through pre-activation using normalization layers and ReLU functions before convolutions; the Large Receptive Field Module, which enlarges the spatial field of view to better capture spatial dependencies and improve keypoint localization; and the Multiscale Module, which extracts features at different resolutions to handle varying scales of body parts. Additionally, intermediate supervision is employed by adding auxiliary losses at multiple stages, allowing the network to refine its predictions progressively and learn more robust representations. The approach is trained and validated using well-established pose estimation datasets like Leeds Sports Pose (LSP) and MPII Human Pose, which include diverse, annotated sports-related postures. The experimental results demonstrate that this multi-module CNN architecture effectively improves the accuracy of posture detection and can be integrated into intelligent training guidance systems to correct physical movements and enhance athletic performance.

Tukino et al explores the use of deep learning techniques, specifically Convolutional Neural Networks (CNNs), for classifying badminton poses with a focus on key techniques such as the service and smash. The methodology in **[4]** involves utilizing Blaze Pose, a lightweight and efficient pose detection model, along with Google's Media pipe Pose Solution to detect and track key body landmarks from video data of badminton players. These tools allow for accurate pose estimation by extracting critical joint coordinates that serve as input features for the classification task. The CNN model is trained using supervised learning on a labeled dataset containing images of various badminton techniques. During training, the model learns to map visual features and pose data to specific action classes. The data processing pipeline includes preprocessing of video frames, key point extraction using Blaze Pose, and feeding this structured data into the CNN for training. To evaluate performance, the study uses classification accuracy as the primary metric, with the model achieving high accuracy rates ranging from 80% to 90%, showcasing its effectiveness in recognizing and categorizing player movements. Overall, the paper presents a robust framework for integrating pose estimation with deep learning to enhance physical activity analysis and technical classification in badminton.

Siddiqui *et al.* focused on classifying cricket batting strokes using Human Pose Estimation (HPE) and machine learning techniques on a custom video dataset comprising eight distinct strokes **[5]**. Using Google's MediaPipe framework, 17 key body landmarks were extracted from video frames and utilized as input features for training multiple classifiers, including Random Forest (RF), Support Vector Machine (SVM), *k*-Nearest Neighbors (KNN), and a Long Short-Term Memory (LSTM) model. Among these, the RF model achieved the highest performance, attaining 99.77% accuracy on the test set and 95.0% accuracy under 10-fold cross-validation, demonstrating strong potential for automated coaching feedback. However, the study's generalizability remains limited due to dataset specificity, with variations in players, lighting conditions, and real-match scenarios requiring further validation. Moreover, the analysis was restricted to kinematic features, excluding contextual factors such as ball trajectory. The authors suggested real-time implementation and the exploration of more advanced deep learning architectures as directions for future research.

Bhosale *et al.* presented a system for yoga pose detection and correction using computer vision techniques **[6]**. The proposed approach employed PoseNet to extract key body landmarks from webcam input and utilized a *k*-Nearest Neighbors (KNN) classifier to recognize one of five trained yoga poses, Goddess, Mountain, Plank, Tree, and Warrior III. The system provided real-time visual feedback by displaying an instructor video corresponding to the detected pose, thereby assisting users in correcting their posture. The model achieved a classification accuracy of 98.51%. However, the system was limited to detecting only five specific yoga poses, posing challenges for scalability to a broader range of asanas. The authors noted that expanding the dataset to include a greater variety of poses and environments could enhance robustness. Additionally, pose estimation accuracy may degrade in cases involving overlapping individuals or occluded body parts. Future work aims to incorporate more complex poses and develop a mobile-based application for self-training.

Two primary technological approaches for posture and movement analysis, sensor-based and vision-based systems, have been discussed by **R. Harini *et al.* [7].** Sensor-based methods employ wearable devices such as inertial measurement units (IMUs) and force sensors, offering high accuracy but limited practicality due to cost, setup complexity, and the need for additional equipment. In contrast, vision-based systems have gained prominence owing to their accessibility and reliance on standard RGB or depth cameras. Frameworks such as OpenPose and MediaPipe enable efficient 2D and 3D pose estimation; however, solutions using depth sensors like Microsoft Kinect remain constrained by hardware dependency. Deep learning architectures, including convolutional and recurrent neural networks, have demonstrated notable improvements in posture recognition accuracy **[7]**. Nevertheless, challenges persist in achieving real-time performance, managing large dataset requirements, and ensuring generalization across diverse users and environments. Although several systems provide real-time feedback through rule-based or classification methods, few offer adaptive and interactive correction during exercises. This underscores a key research gap, the need for a lightweight, real-time, camera-only system capable of delivering accurate and immediate feedback without reliance on external sensors or specialized hardware.

Farida Asriani *et al.* proposed a highly effective hybrid deep learning model designed to enhance the accuracy of badminton action recognition, a task made challenging by subtle stroke variations and the need to capture complex temporal dynamics of body movements. The study introduces a dual-stream framework that integrates complementary data modalities from video clips of PBSI-registered athletes **[10]**. Initially, ten representative keyframes are selected from each video using linear interpolation. These frames are processed in parallel, where a pre-trained Vision Transformer (ViT) model (*google/vit-base-patch16-224-in21k*) extracts 7,680 features from raw RGB frames to encode global contextual information, while Google's MediaPipe framework derives 3D coordinates of 33 body landmarks, generating 990 skeletal features that accurately capture motion and posture. The two feature sets are concatenated into a single composite vector and input into a Long Short-Term Memory (LSTM) network. The LSTM classifier is utilized for its ability to model temporal dependencies within sequential data, effectively interpreting entire stroke sequences over time. Experimental results validate the strength of this hybrid strategy, with the model achieving a peak recognition accuracy of 98.68%. This significantly outperforms unimodal baselines, including the best skeleton-only model (95.26%) and ViT-only model (96.84%). The study concludes that fusing structural skeletal data with contextual visual information yields a more comprehensive representation of player actions, establishing a new state-of-the-art benchmark and offering valuable insights for future automated sports analytics and intelligent coaching systems.

The accurate localization of human joints from visual data serves as a cornerstone of modern biomechanical and motion analysis. Foundational research in this field, such as the work by **Cao *et al.*** on OpenPose, established robust methodologies for real-time, multi-person 2D pose estimation, as described in **[11]**. Their proposed architecture, which utilizes Part Affinity Fields (PAFs) and Confidence Maps, provided a seminal framework for associating skeletal keypoints in visually complex environments, such as multi-player sports scenes. This research laid the foundation for accessible, high-performance frameworks like MediaPipe, introduced by **Lugaresi *et al.***, whose work in **[12]** details a modular and cross-platform perception pipeline optimized for real-time inference on resource-constrained devices.

Within the MediaPipe ecosystem, the BlazePose model proposed by **Bazarevsky *et al.*** represents a significant advancement in on-device pose tracking. As presented in **[13]**, the architecture employs a two-stage vision pipeline that integrates a lightweight detector with a regression network capable of predicting 33 distinct body keypoints, aligning well with real-time feedback applications in sports and fitness contexts. The historical foundation for such regression-based pose estimation approaches was established by Toshev and Szegedy, who introduced DeepPose as one of the earliest deep learning frameworks for human pose estimation. As stated in **[14]**, their work pioneered the concept of directly regressing Cartesian joint coordinates from image pixels, marking a paradigm shift from traditional feature-engineered methods to end-to-end deep neural networks.

The classification of dynamic actions, such as badminton strokes, requires architectures capable of interpreting spatio-temporal information, as discussed in **[15].** The seminal work on Long-term Recurrent Convolutional Networks (LRCN) by **Donahue *et al.*,** provides a strong rationale for our model selection. The LRCN architecture effectively combines

Convolutional Neural Networks (CNNs), which extract spatial features from individual frames, with Long Short-Term Memory (LSTM) networks, which model temporal dependencies across sequences. An influential alternative was proposed by Simonyan and Zisserman in their Two-Stream Convolutional Networks, as noted in **[16]**. This architecture processes a spatial stream (static frames) and a temporal stream (optical flow) in parallel, demonstrating that explicit motion modeling is critical for high-performance action recognition.

Further advancing the field, Carreira and Zisserman introduced the Inflated 3D ConvNet (I3D), which extends 2D convolutional filters into the temporal dimension to create a unified 3D network, as highlighted in **[17]**. This approach allows for the simultaneous learning of spatial and temporal features. The development of I3D, together with the introduction of the large-scale Kinetics dataset, marked a significant milestone in video understanding and provides a powerful alternative to recurrent architectures for action classification tasks.

The translation of these computer vision techniques into applied sports science is an active research area. The study by **Harini *et al*.** on real-time squat correction, in **[18]**, serves as a direct methodological parallel to our project. They implemented an end-to-end system using MediaPipe for keypoint extraction, a deep learning model for classification, and a rule-based engine for providing immediate corrective feedback, thereby validating our proposed architecture. A common challenge in sports analysis is robust player identification. As mentioned in **[19]**, **Ribeiro *et al.*** conducted a systematic review on athlete re-identification in broadcast videos, exploring techniques beyond simplistic ROI filters, such as appearance-based feature matching and motion signature analysis, to maintain player identities in dynamic scenes.

While focused on a different domain, the review by **Whiteside *et al.* [20]** provides a comprehensive overview of computer vision applications in tennis, encompassing player tracking, stroke analysis, and ball trajectory estimation, with several of these challenges being directly transferable to badminton. For analyses that demand higher anatomical precision, the work by **Güler *et al.* [21]** on DensePose introduces a computer vision method for mapping the full 3D surface of the human body from a 2D image, enabling more detailed evaluation of body segment rotation and muscle activation , a direction highly relevant for advanced biomechanical feedback systems.

The performance of supervised learning models heavily depends on the availability of high-quality, domain-specific datasets; for instance, **Wang *et al.* [22]** contribute a public, annotated dataset for fine-grained badminton action recognition, emphasizing methodological rigor during dataset creation. The scientific legitimacy of vision-based biomechanical assessment has been validated by **Stenum and Ishøi** as shown in **[23]**, whose work demonstrates that monocular 3D pose estimation can accurately compute kinematic parameters such as joint angles , reinforcing the basis of our corrective feedback mechanism.

A practical demonstration of such biomechanical modeling is evident in the study by **Nakashima and Murai as in [24]**, who employed deep learning to derive 3D pose data for automated baseball pitching analysis, computing metrics like joint angular velocities to quantify motion efficiency. Finally, **Liu *et al.*** in **[25]** present a deep metric learning framework that parallels our Key Similarity Index (KSI) approach; while our metric is hand-engineered for motion comparison, their method uses neural networks to learn optimal similarity functions from data, suggesting a promising future path for refining our comparison engine.

2.2 Literature Survey Analysis

While current research finds considerable advancement in the use of computer vision to human pose estimation and action recognition in a wide range of applications, including sports such as badminton, cricket, and yoga, there remains some scope, and, in particular, for badminton training software. The majority of research targets the movement classification or pose estimation, while fewer examine the provision of detailed, real-time corrective feedback commensurate with the dynamic, complex nature of the badminton shot.

Even correction-based systems, such as the reviewed yoga app comparing user pose to an instructor, are bounded in scope (e.g., number of poses 4) and may not be entirely transferable to correcting complex sports movements. Most importantly, there does not seem to be a specialist system designed not only for detection or simple correction, but as complete teaching aids that can teach novice learners to learn basic badminton shots or aid novice players to systematically improve their skills through targeted feedback and potentially mobile accessibility. Thus, there is considerable potential to create an application combining accurate posture detection with specific, actionable corrective advice tailored to badminton, both as a skills development tool for novices and a refinement tool for amateurs

2.2.1 Consolidated Research Gap

1: Most existing research focuses on pose estimation or movement classification, with limited emphasis on real-time corrective feedback.

2: Current correction-based systems (e.g., yoga applications) have limited scope in terms of the number of poses and are unsuitable for complex, fast-paced sports like badminton.

3: There is a lack of complete teaching aids that systematically teach or improve novice players' badminton skills.

4: Limited mobile accessibility in existing sports training systems restricts user convenience and flexibility.

5: Most research models are limited to a small number of shot types and movement patterns in badminton analysis.

6: Many posture detection systems rely solely on client-side or server-side processing, leading to performance or latency issues.

3 Problem Definition

Badminton players often face challenges with improper posture and technique during training, which not only hampers their skill development but also increases the risk of injury. The lack of accessible, automated systems capable of providing real-time, objective feedback makes it difficult to correct these issues effectively. To address this, we propose the development of an AI-driven computer vision system that analyzes player movements in real time and delivers actionable, data-based feedback.

By automatically detecting incorrect postures through pose estimation and offering instant visual or textual guidance, the system helps players refine their form efficiently. Additionally, it tracks performance over time to enable personalized training and continuous improvement. This solution bridges the gap between subjective coaching and precise, consistent correction, hence reducing human error, providing timely feedback, and ultimately making badminton training more effective and scalable.

4 Proposed System

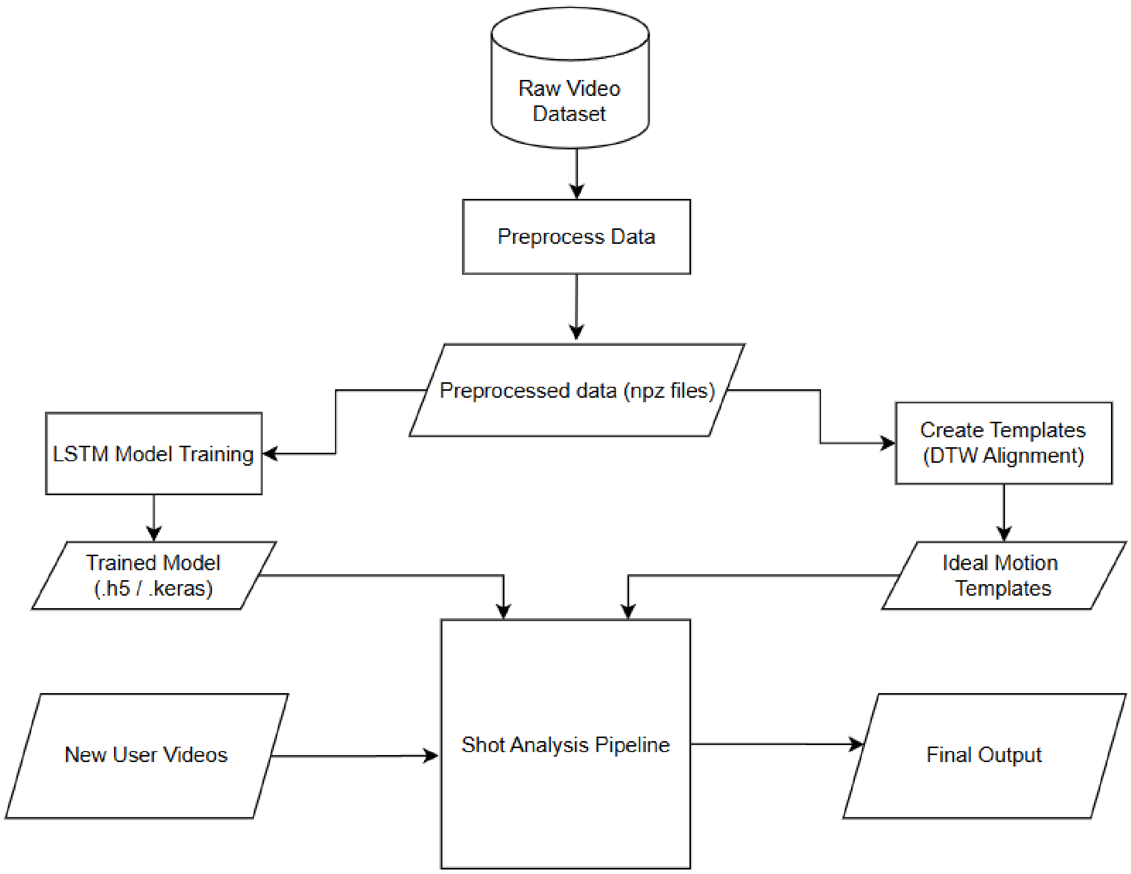


Figure 4.1 : Overall Approach

This section details the journey of a single video frame from raw pixel data to an analyzed piece of coaching advice.

Step 1: Player Detection and Isolation

The primary challenge on a badminton court is ambiguity. To provide feedback to the correct person, we must first reliably isolate them from their opponent and any other individuals in the frame.

- **Method in Detail:** We employ a Region of Interest (ROI) filter, which acts as a spatial gate. We configure the system with a vertical threshold (e.g., $y < 0.5$), instructing it to only consider poses whose vertical center is in the top half of the screen. MediaPipe's Pose solution is optimized to find a single prominent person; our ROI acts as a crucial secondary check to ensure it has locked onto the correct target player and discards any detections of the closer, foreground player. This simple but effective heuristic is vital for automating the analysis of real-world game footage.

Step 2: 3D Keypoint Extraction

Simply knowing a player's 2D position on the screen is insufficient for genuine biomechanical analysis. To understand limb rotation, depth, and true angles, we require 3D data.

- **Method in Detail:** For each valid frame, MediaPipe's BlazePose model is used. It infers 33 key body landmarks. Crucially, we use the pose_world_landmarks output, which provides the joint coordinates in a real-world 3D space, measured in meters from the center of the hips. This is a significant advantage over 2D coordinates because it is viewpoint-invariant. An elbow bent at 90 degrees will be calculated as such regardless of whether the player is facing the camera, is sideways, or is at an angle. This ensures the analysis is robust and accurate. The output is a sequence of vectors, [frames, 33 joints, 3 coordinates (x,y,z)], which forms the raw data for our subsequent analysis.

Step 3: Shot Classification

Providing feedback for a "smash" when the player was attempting a "drop shot" would be useless. Therefore, contextual understanding of the action is paramount.

- **Method in Detail:** The entire sequence of 3D landmark data is fed into a pre-trained Long Short-Term Memory (LSTM) network. An LSTM was chosen over simpler models (like Random Forest) or standard CNNs because it is specifically designed to recognize patterns in sequential data. It maintains an internal "memory" that allows it to understand the relationship between the pose in the current frame and the poses in previous frames. This enables it to learn the unique temporal "signature" of each badminton shot, for example, the rapid, explosive arm movement of a smash versus the slower, more controlled motion of a lift.

Step 4: Posture Analysis and Correction (KSI)

This is the core of the project, where the numerical data is translated into coaching insights.

- **Method in Detail:** Once the LSTM classifies the shot as, for example, a "forehand drive," the system retrieves a pre-recorded, high-quality "expert" sequence for that same shot. The user's motion and the expert's motion are then fed into the Kinetic Similarity Index (KSI) engine. This engine doesn't just say "correct" or "incorrect"; it provides a nuanced score across several dimensions of the movement. A rule-based system then maps these scores to specific, pre-defined feedback statements. For instance, a low score in the acceleration component of the KSI will trigger a message focused on generating more power.

4.2 Techniques / Methodology

4.2.1 The KSI Algorithm: A Deeper Dive

The KSI formula is designed to mimic how an expert coach would analyze a movement: by looking at form, speed, and power as separate but interconnected components.

Equation (4.1):

KSI = wp × Spose + wv × Svelocity + wa × Sacceleration

Step 1: Temporal Alignment (DTW)

- Intuition: Imagine placing two audio tracks of the same song next to each other, but one is played slightly faster. If you compare them second-by-second, they will quickly fall out of sync. Dynamic Time Warping (DTW) is like a smart audio engineer who stretches and compresses the tracks to perfectly align the beats before comparing them. It finds the optimal frame-to-frame mapping between the user's motion and the expert's, ensuring we are always comparing corresponding parts of the swing.

Step 2: Calculate Postural Alignment (S_pose)

- Intuition: This component answers the question: "Is the player making the right 'shapes' with their body?" It focuses purely on the angles of the joints, ignoring speed or power.
- Formula & Rationale: We use Cosine Similarity to compare the vectors of the user's and

expert's limbs.

- The formula, Spose

Equation (4.2):

$$S_{pose} = \frac{1}{N}\sum_{t=1}^{N}\left(\frac{V_{user,t}\cdot V_{expert,t}}{||V_{user,t}||\,||V_{expert,t}||}\right)$$

works by isolating the cosine of the angle between two vectors. This brilliantly makes the comparison independent of the player's actual limb length, ensuring a tall player and a short player can both be judged against the same ideal form.

Step 3: Calculate Velocity Coherence (S_velocity)

- Intuition: This component answers: "Is the player moving their limbs at the right speed and in the right direction?" A player could have perfect form but swing far too slowly.
- Formula & Rationale: The formula,

Equation (4.3):

$$S_{mag,t} = \exp\left(-\alpha(\Delta speed)^2\right) = e^{-\alpha(\Delta speed)^2}$$

is a product of two parts. The Cosine Similarity part checks if the user's wrist is moving in the same *direction* as the expert's. The Gaussian Function (e^{-x^2}) part creates a "bell curve" score for speed. It gives a perfect score of 1 if the speeds match exactly, and the score gracefully decreases the larger the difference in speed, penalizing large deviations more heavily than small ones.

Step 4: Calculate Acceleration Profile (S_acceleration)

- Intuition: This component answers: "Is the player generating force at the right moment?"

This is the key to explosiveness and power. A badminton smash isn't about moving the arm fast the whole time; it's about a rapid acceleration just before impact.

- Formula & Rationale: The formula

Equation (4.4):

$$S_{acceleration} = \frac{1}{N}\sum_{t=1}^{N} \exp\left(-\beta\left(\|a_{user,t}\| - \|a_{expert,t}\|\right)^2\right)$$

again uses a Gaussian Function. It compares the magnitude of the user's wrist acceleration to the expert's at every moment. It will give a high score only if the user generates that crucial burst of acceleration at the exact same phase of the swing as the expert.

Step 5: Calculate Final KSI Score

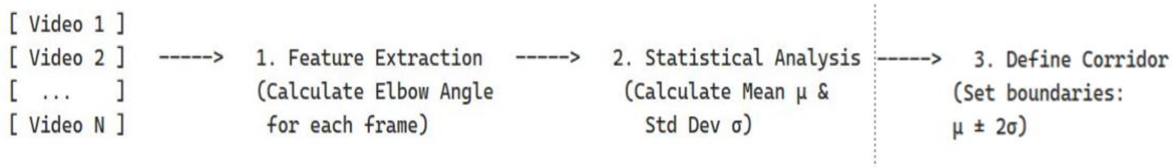
- Intuition: This is the final report card. It combines the individual grades for form, speed, and power into a single, overall score.
- Formula & Rationale: The weighted average,
  KSI=(wp·Spose)+(wv·Svelocity)+(wa·Sacceleration),
  allows for flexibility. For a delicate net shot, you might increase the weight for Spose (form is critical). For a powerful smash, you might increase the weights for Svelocity and Sacceleration (speed and power are key). This makes the KSI a highly adaptable tool for analyzing different types of shots.

4.2.2 Kinematic Thresholding: The Validation Model

This approach defines a "corridor of correctness" from training data and then validates a user's motion against it in real-time.

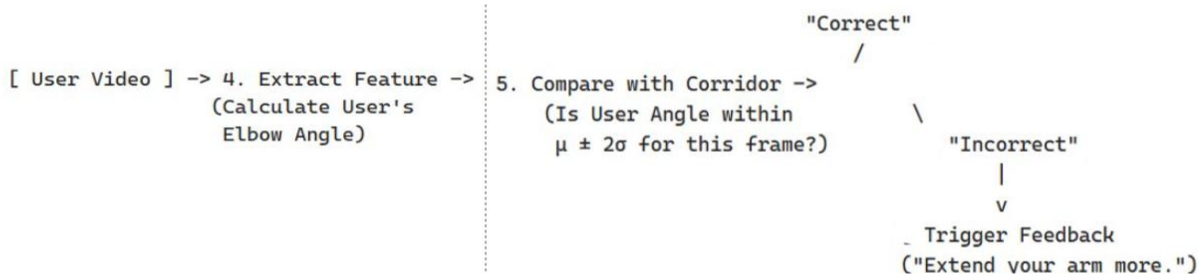
Phase 1: Offline Model Building (From Training Data)

Goal: To define the statistical boundaries of a "correct" movement.



Phase 2: Real-Time User Analysis

Goal: To check if the user's movement stays within the defined corridor



5 Result

5.1 Dataset Description

This dataset contains video data of various badminton strokes, organized into distinct folders corresponding to specific actions:

- backhand_drive
- backhand_net_shot
- forehand_clear
- forehand_drive
- forehand_lift
- forehand_net_shot

Each folder contains videos capturing human subjects performing the specific stroke from multiple angles and under consistent conditions, aiding in effective action recognition and pose analysis.
The total size of the dataset is 1.32 GB, making it manageable for local development and training with modest hardware (Using RTX 4070).

Purpose and Relevance to the Project:
This dataset is particularly beneficial for our posture correction and estimation model due to the following reasons:

1. Diverse Motion Capture: It includes a variety of badminton-specific movements, ideal for training models to understand and evaluate sports posture in dynamic settings.
2. Class-Labeled Data: The clear categorization by stroke type simplifies supervised learning and enables accurate classification.
3. Video-Based Input: The dataset allows us to extract frame-wise pose keypoints, which

are crucial for estimating joint angles and correcting posture in real-time.

4. Realistic Action Execution: The dataset mimics real-world scenarios, helping to generalize our model beyond lab conditions.

Proposed Data Split:

To train, validate, and evaluate the model effectively, the dataset will be divided as follows:

- Training Set (70%): Used for model learning and optimization**.**
- Validation Set (15%): Used to tune model parameters and prevent overfitting.
- Test Set (15%): Used for final evaluation of the model's performance.

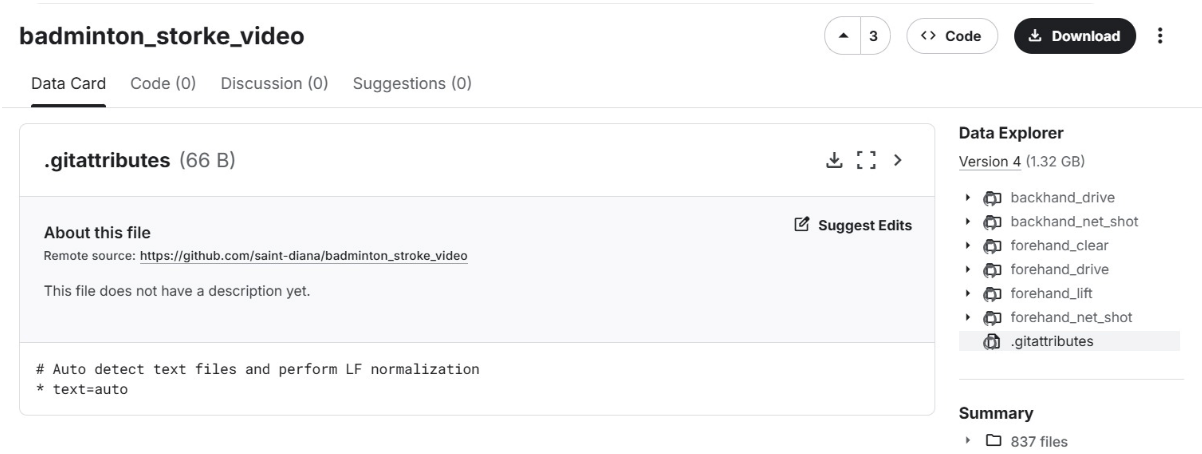
This split ensures a balanced distribution of data for robust model training and reliable performance evaluation.

License and Usage:

This dataset is published under the Creative Commons License (CC BY 4.0) on Kaggle. This license allows:

- Sharing – Copy and redistribute the material in any medium or format.
- Adaptation – Remix, transform, and build upon the material for any purpose, even commercially.

With the condition that appropriate credit is given to the original dataset creators and source. Therefore, this dataset is suitable for academic and development purposes under open research standards.



5.2 Implementation Details

The implementation of the system is divided into five primary stages: Data Acquisition and Preprocessing, Pose Normalization, Model Architecture and Training, Real-time Inference, and Posture Correction. The entire pipeline was developed in Python, leveraging libraries such as OpenCV for video processing, MediaPipe for pose estimation, and TensorFlow/Keras for building and training the deep learning model.

5.2.1 Data Acquisition and Preprocessing

The initial stage focuses on converting raw video footage into a format suitable for the neural network.

1. **3D Landmark Extraction**: The system utilizes Google's MediaPipe Pose library, specifically the pose_world_landmarks functionality. For each frame of a video, MediaPipe estimates the 3D coordinates of 33 key body joints. These world coordinates are provided in meters and are relative to a coordinate system originating at the center of the hips, making them robust to camera distance. This process is encapsulated in the extract_3d_landmarks_from_video function._
2. **Temporal Segmentation (Sliding Window)**: A single badminton shot is a dynamic action that unfolds over a sequence of frames. To capture this temporal information and augment the dataset, a sliding window approach is employed. A fixed-size window (e.g., 40 frames) slides across the extracted landmark sequence with a defined stride. This method generates multiple training samples from a single video, capturing the shot's execution from various starting points._

5.2.2 Viewpoint Invariance via Pose Normalization

A significant challenge in action recognition is viewpoint variance; the same action can appear drastically different depending on the camera's angle and position. To address this, a rigorous pose normalization pipeline, implemented in the normalize_sequence function, is applied to every windowed sequence before training. This ensures the model learns the intrinsic motion of the shot, not the camera's perspective. The process involves three steps:

1. **Centering (Translation)**: The entire skeleton is translated so that its center, defined as the midpoint of the left and right hip joints (landmarks 23 and 24), is at the origin (0, 0, 0). This removes positional bias.
2. **Alignment (Rotation)**: A new, person-centric coordinate system is established. The primary "up" vector (Y-axis) is defined by the line from the hip center to the shoulder center. The "right" vector (X-axis) is defined by the line connecting the two shoulders, made orthogonal to the "up" vector. The "forward" vector (Z-axis) is calculated using the cross-product of the other two. A rotation matrix is then computed to align this person-centric system with the standard XYZ axes. This critical step ensures that the player always appears upright in the data, regardless of their orientation to the camera.
3. **Scaling**: To ensure scale invariance (i.e., the player's distance from the camera does not affect the data), the entire normalized skeleton is scaled relative to a consistent anatomical feature. The length of the spine (the distance from the hip center to the shoulder center) is used as the scaling factor.

This multi-step normalization makes the model significantly more robust and capable of generalizing from the training data.

5.2.3 Model Architecture and Training

The core of the classification system is a Long Short-Term Memory (LSTM) based neural network, designed to recognize patterns in sequential data.

1. **Model Architecture**: The model takes a sequence of flattened and normalized 3D keypoints as input. The architecture consists of stacked LSTM layers with relu activation, which are adept at learning temporal dependencies. Dropout layers are strategically placed between the LSTM layers to prevent overfitting by randomly deactivating neurons during training. The network concludes with Dense layers,

culminating in a softmax activation function that outputs a probability distribution across the different shot classes._

2. **Training**: The preprocessed and normalized dataset is split into training and validation sets. The model is compiled using the Adam optimizer and the categorical_crossentropy loss function, which is standard for multi-class classification. The model is trained for a set number of epochs, and the version with the highest validation accuracy is saved for inference._

5.2.4 Real-time Inference

The final implementation provides real-time feedback on a live video stream (from a file or webcam).

1. **Sequence Buffering**: The system maintains a sliding buffer (a deque) of the most recent 40 frames of detected landmarks._
2. **On-the-fly Preprocessing**: As new frames are captured, the landmarks are added to the buffer. Once the buffer is full, the entire sequence is passed through the same normalize_sequence function used during training. This ensures that the data fed to the model for prediction is in the exact same format as the training data._
3. **Prediction Smoothing**: To prevent the predicted class from flickering rapidly, a second buffer is used to store the last several prediction probability vectors. The final displayed prediction is based on the average of these probabilities, resulting in a more stable and reliable output._

5.2.5 Posture Correction and Feedback Generation

Beyond classifying the shot, the system is designed to provide corrective feedback by comparing the user's motion to an ideal template.

1. **Motion Template Generation**: For each shot class, an "ideal" motion template is created. This is achieved by taking multiple high-quality examples of a shot (e.g., from expert players), aligning them temporally, and averaging their normalized keypoint sequences. The resulting template represents the biomechanically optimal execution of that shot._
2. **Performance Comparison with DTW**: When a user performs a shot, the system first classifies it. Then, the user's captured and normalized keypoint sequence is compared

against the corresponding ideal motion template. To account for variations in execution speed, Dynamic Time Warping (DTW) is used. DTW is an algorithm that finds the optimal alignment between two temporal sequences, effectively stretching or compressing them to match up corresponding moments in the action.

3. **Error Identification and Feedback**: After DTW alignment, the system calculates the Euclidean distance between the user's keypoints and the template's keypoints for each frame. If the deviation for specific joints (e.g., the elbow, wrist, or shoulder) exceeds a predefined threshold at critical phases of the swing, the system generates targeted, actionable feedback. For example, a large deviation in the elbow joint during the point of contact might trigger a message like, "Elbow movement is inaccurate, ensure you are extending your arm fully." This rule-based feedback mechanism provides immediate, understandable advice for improvement.

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